

Steady-state thermal condition analysis system for heat transfer tubes in i-SMR, and replacement of Inconel690 with composite materials



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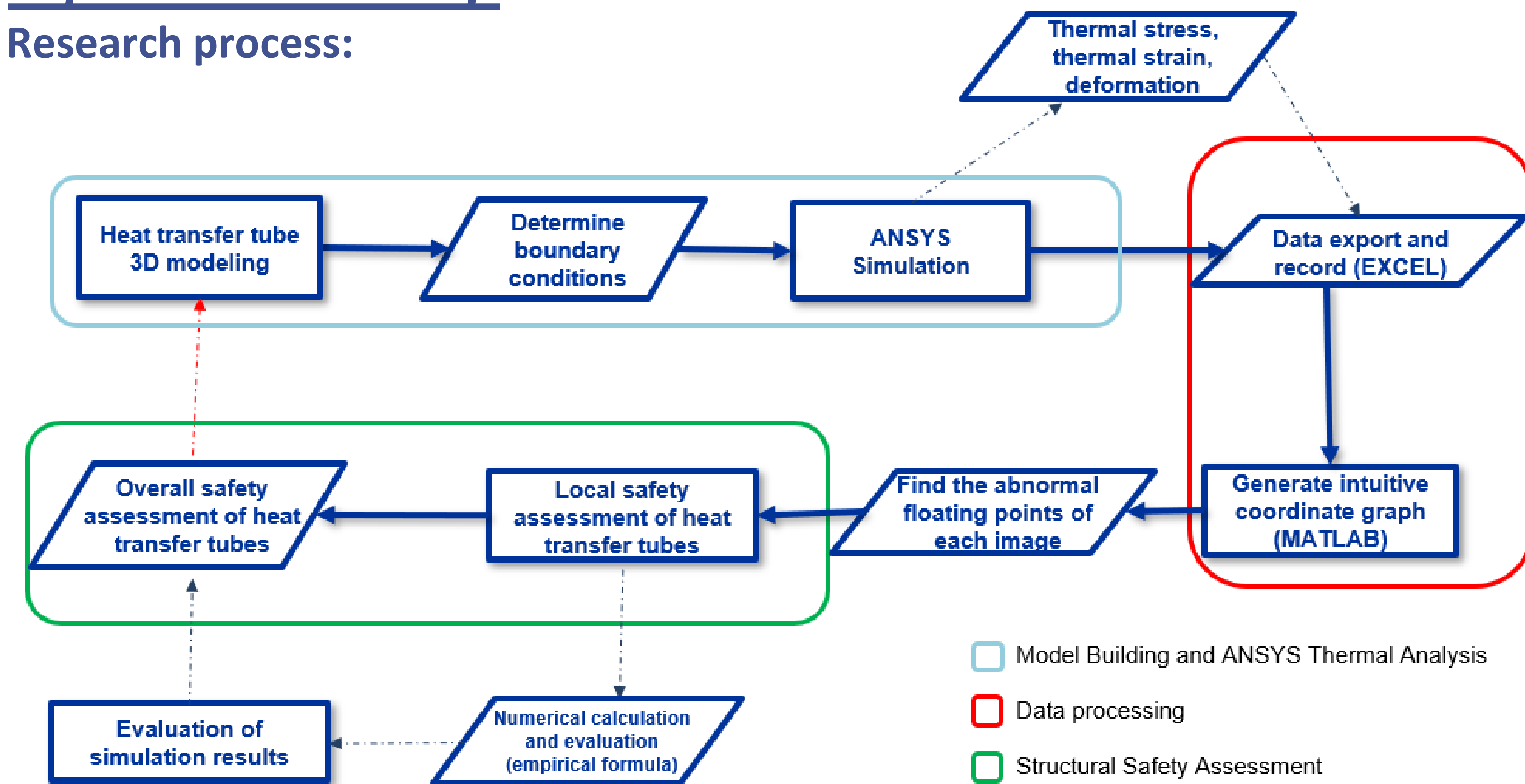
Introduction

- Energy Transition and Carbon Neutrality Background
- Modularity and High Safety of SMRs
- Structural Safety of OTSG Heat Exchanger Tubes
- Performance Limitations of Inconel 690
- Advantages of PI/20%BN Composite Materials as an Alternative

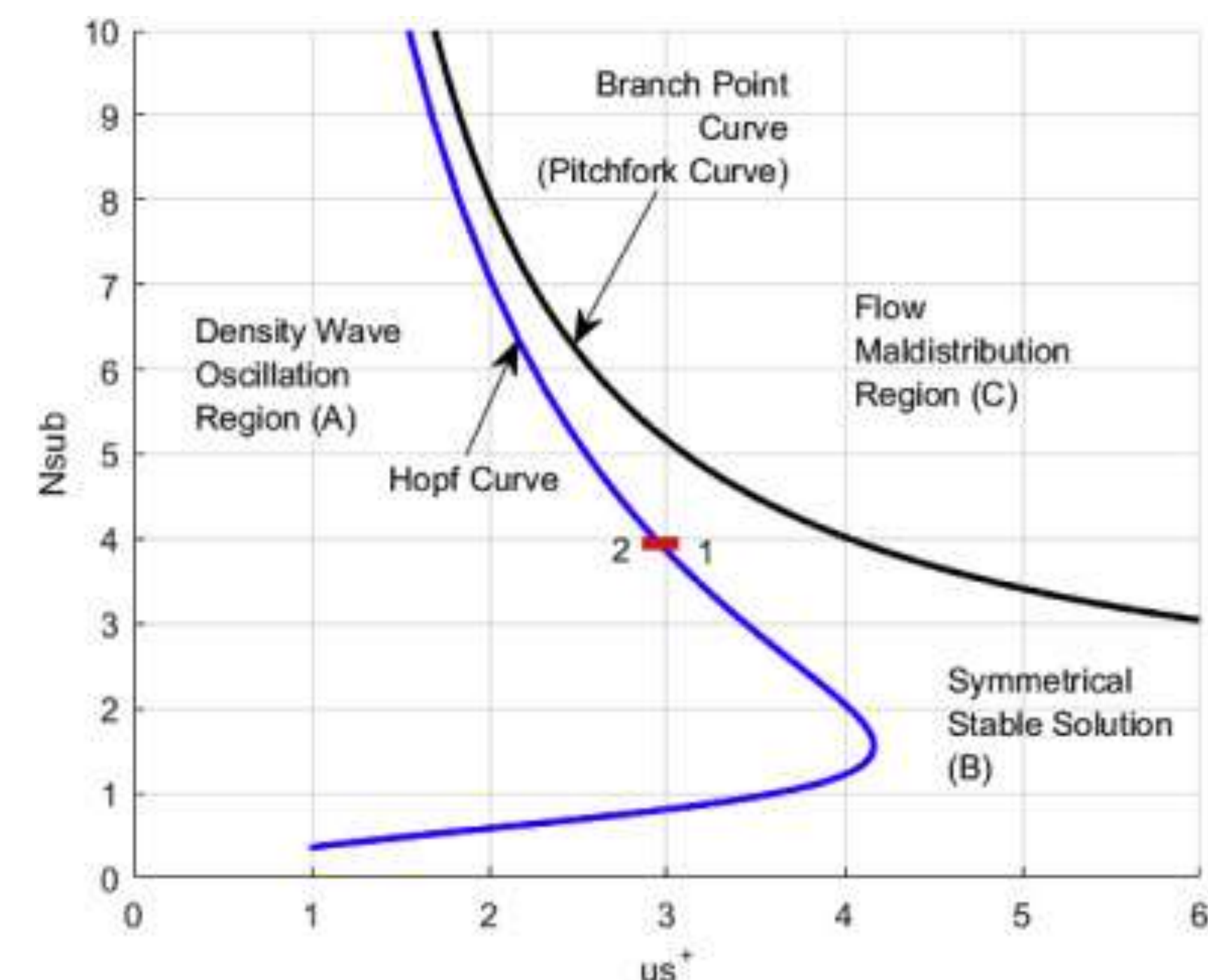


Experimental Setup

Research process:



Challenge : Density Wave Oscillation

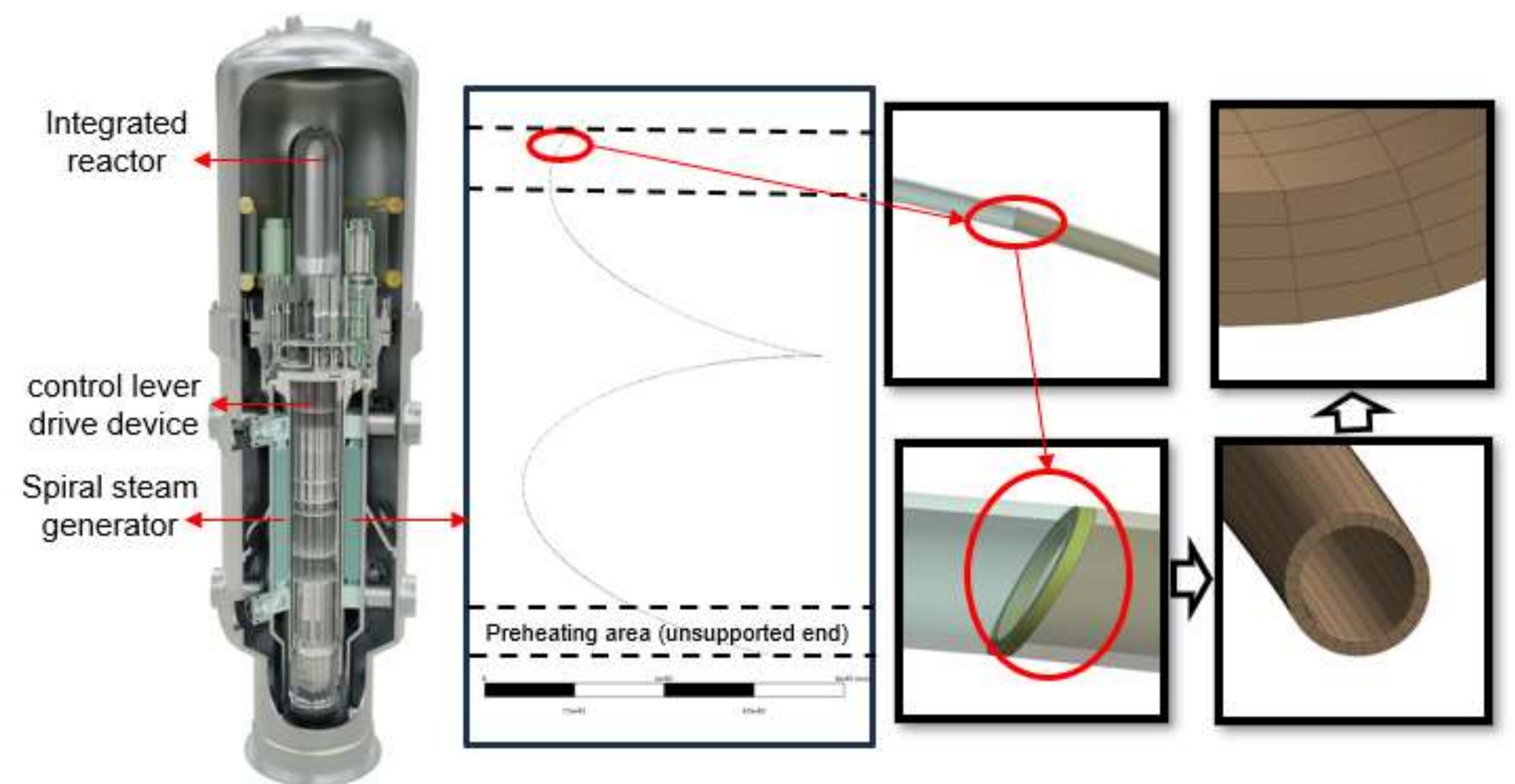


Model :

- A preheating zone is set at both the inlet and outlet, with 0.1 turns, without any support constraints.
- Fixed-end constraints are applied at the inlet and outlet to simulate fixed pipe end boundaries.
- 23 supports are evenly distributed axially in the central region, restricting only z-direction displacement to maintain structural stability.

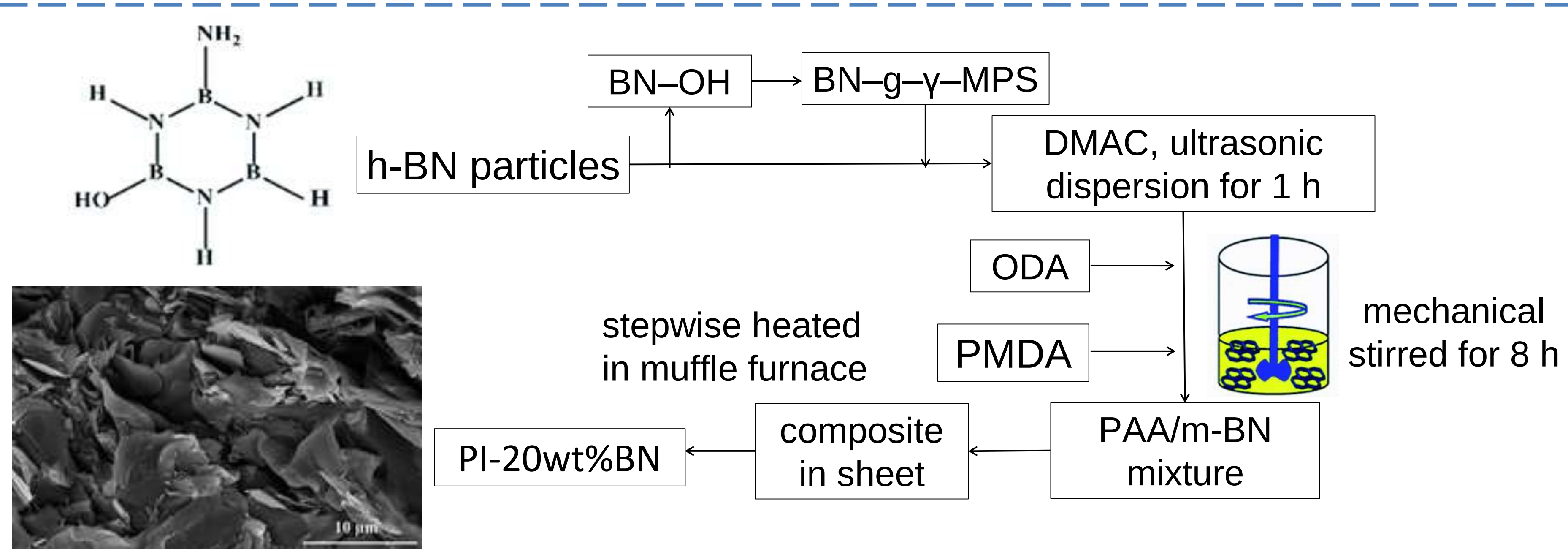
Mesh and analysis settings: :

- Meshing method: Sweep
- Nodes 2,721,390
- Elements 572,880
- Temperature loading: Piecewise temperature setting



Material : PI-20wt%BN

Property	Pure PI	PI-BN	PI-CNT	Improvement
Thermal conductivity (W/ m· K)	0.2	0.7	1.2	↑ 300~500%
Tensile modulus (G Pa)	3.0	4.5	5.2	↑ 60~70%
Dielectric strength (MV/m)	230	300	250	↑ 20~30%



Results

Data processing: :

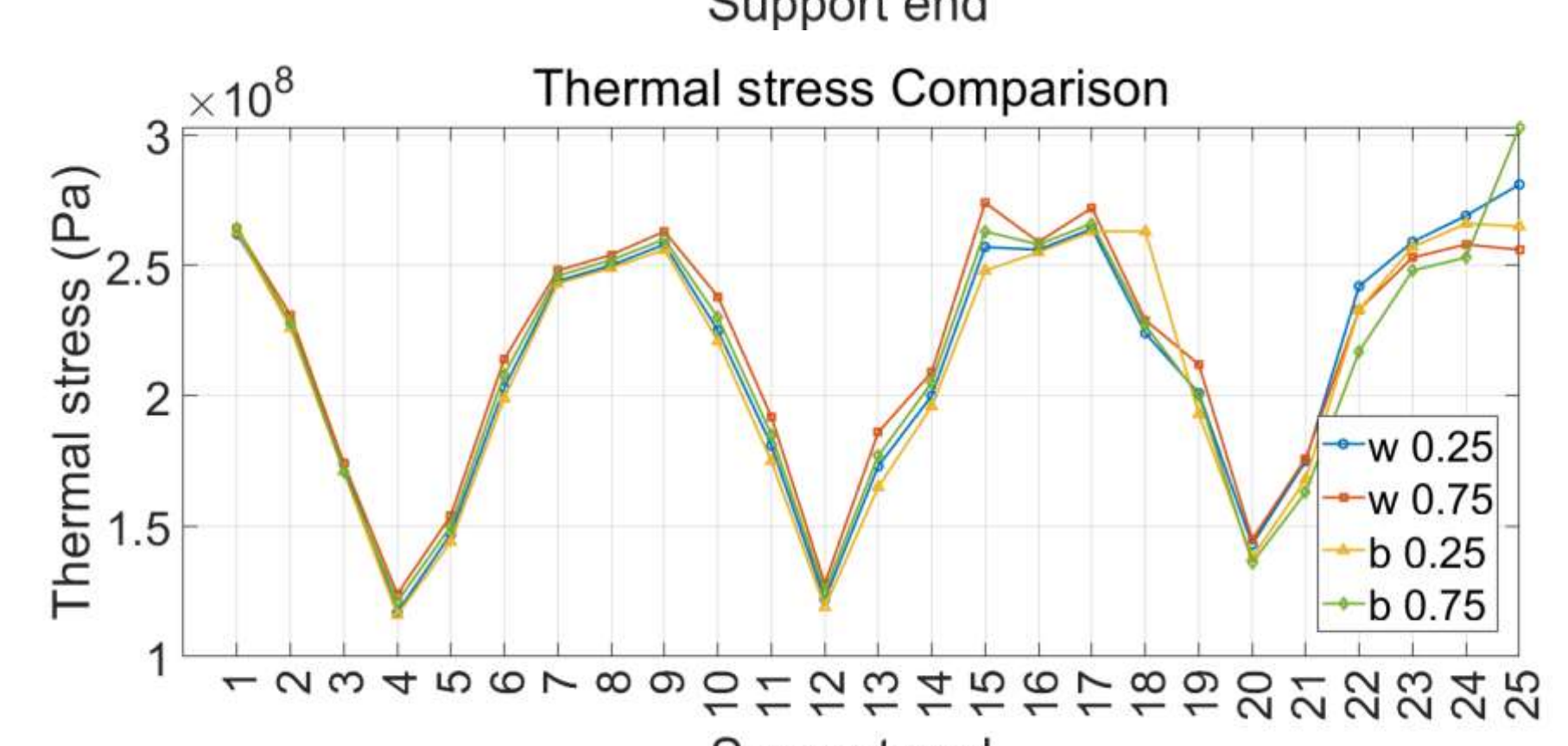
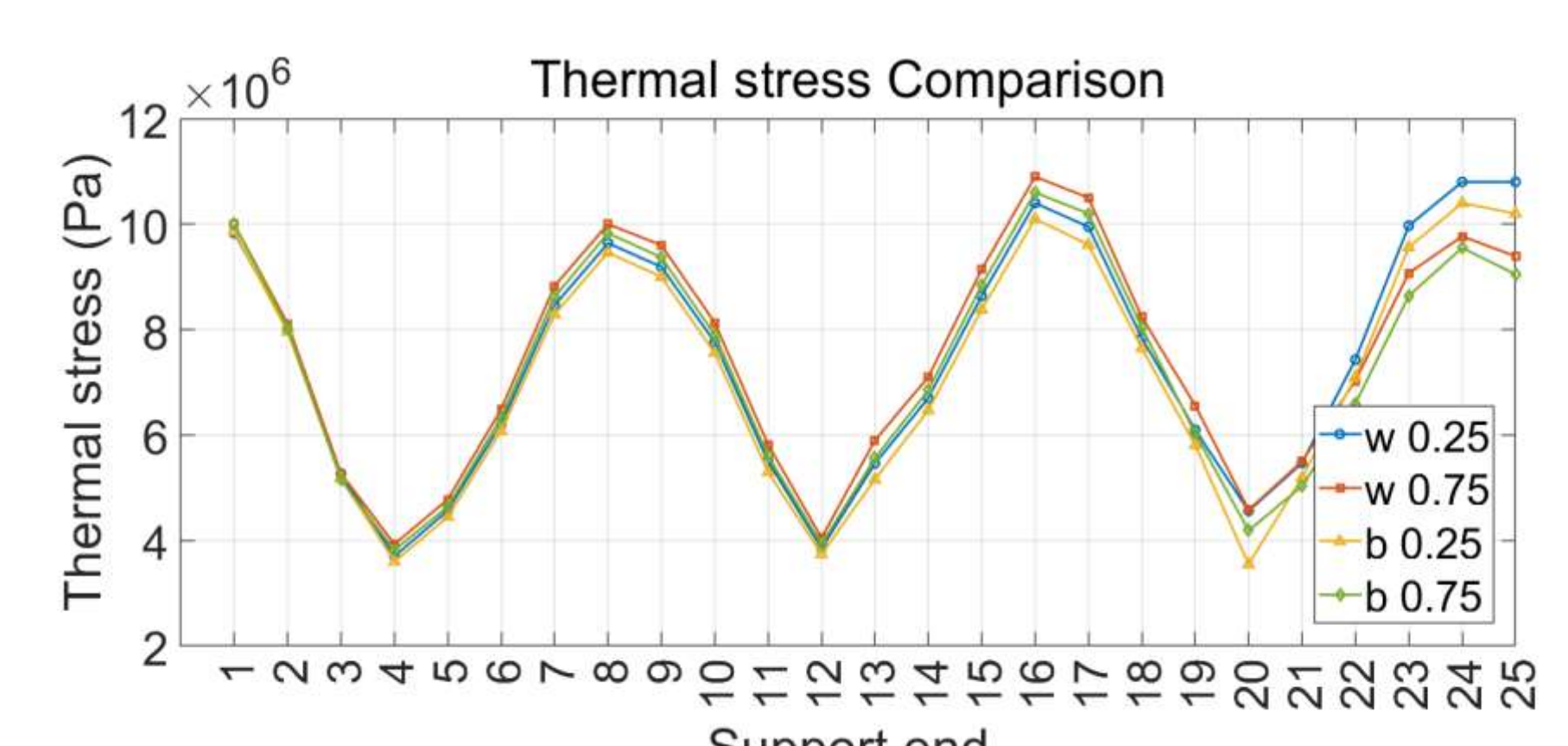
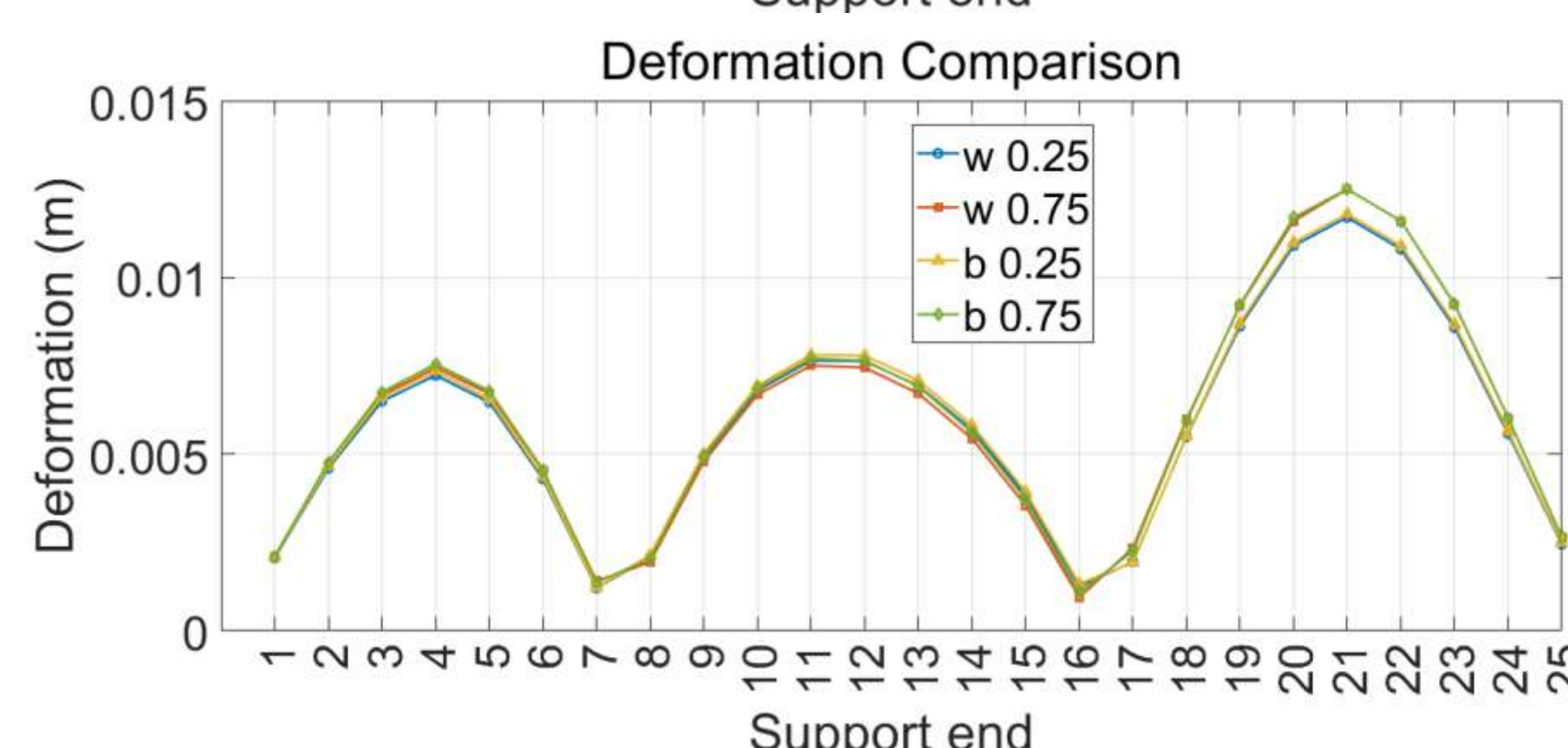
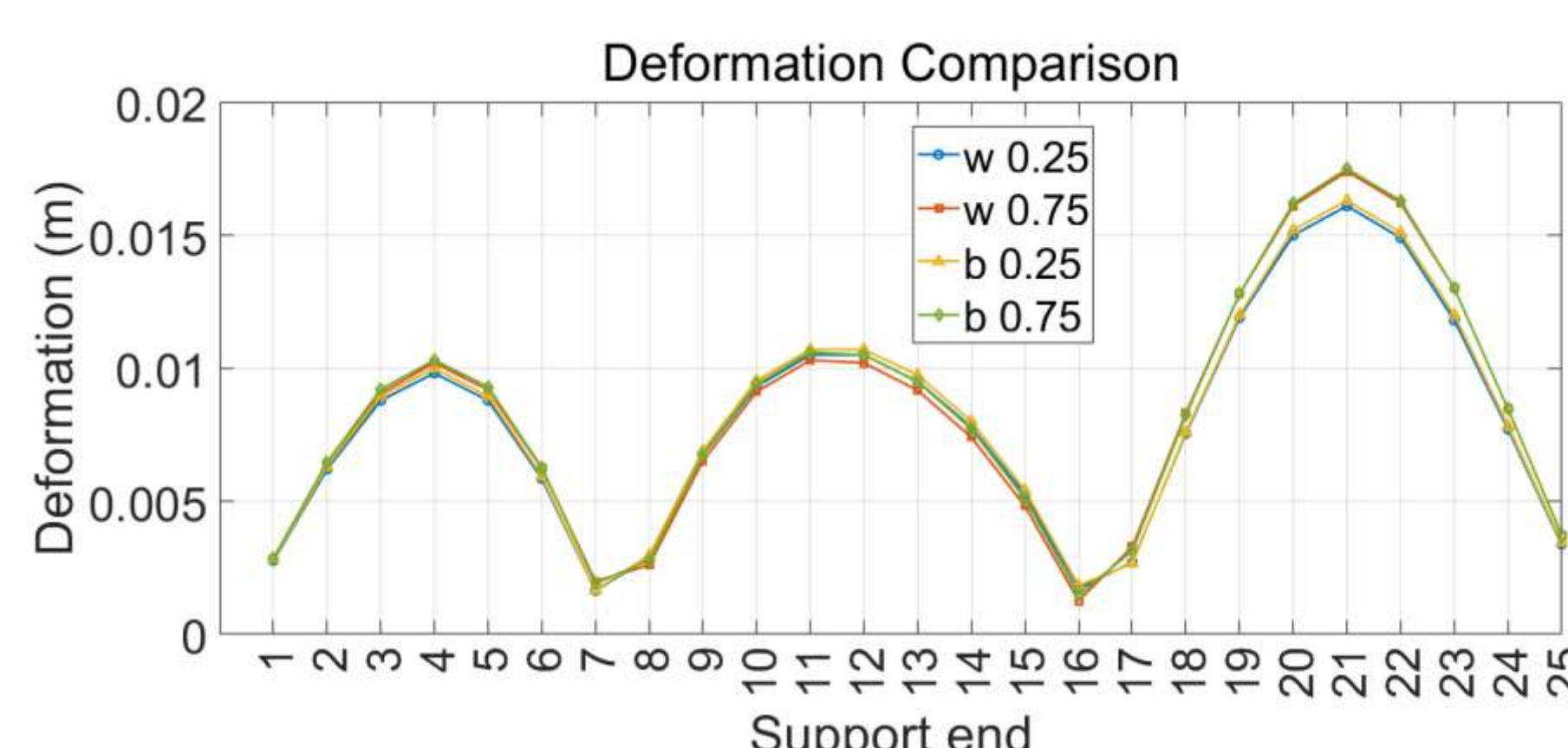
- Export ANSYS calculation results
- Condition the data to an Excel spreadsheet
- Visualize the data using MATLAB
- Display the temperature and stress distribution patterns

Conclusion

- The deformation of the PI material increased slightly but remained uniform.
- Thermal stress was significantly reduced by approximately twenty times.
- This resulted in both lightweighting and improved structural stability.

Future Works

- ANSYS ACP Sandwich Structure Design (PI/Metal Liner Composite Layer) • Reduces deformation and thermal stress, ensuring safe control parameters



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